

Warm Up Exercises: Unit 2, Forces

Prof. Jordan C. Hanson

September 24, 2025

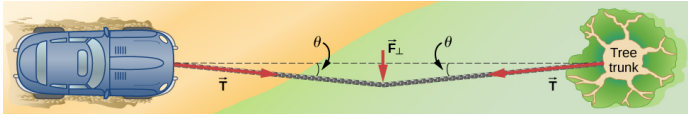


Figure 1: Person pushes in a direction orthogonal to a rope connecting a car and a tree.

3. Consider Fig. 1, in which a rope is tied to a tree and a car stuck in mud. The force is perpendicular to the middle of the rope, $\vec{F}_\perp$. Suppose $\vec{F}_\perp$, the left-pointing tension $\vec{T}$, and right-pointing tension $\vec{T}$ all cancel to yield $\vec{F}_{net} = 0$, show that

$$2T \sin \theta = F_\perp \quad (1)$$

Hint: it helps if you think of the tension vectors as pointing the opposite direction as shown in Fig. 1.

1 Memory Bank

1. $\vec{F} = m\vec{a}$... Newton's 2nd Law
2. $\vec{w} = -mg\hat{j}$... Weight force
3. $v_f^2 = v_i^2 + 2a\Delta x$... Kinematic equation without time

2 Chapter 4 - Forces

1. A rocket of mass m is falling under the influence of gravity, but experiences a thrust force upwards $\vec{F}_t$, making the net force $\vec{F}_{Net} = \vec{F}_t - \vec{w}$. (a) Suppose the velocity of the rocket is $v(t) = (1/2)(k/m)t^2 - gt$, implying that the acceleration is $a = (k/m) - g$. Write an expression for $\vec{F}_t$. (b) If the velocity at $t = 10$ seconds is 50 m s^{-1} , and $m = 150 \text{ kg}$, what is k ? (c) What is the net force, in Newtons, at $t = 10$ seconds?
4. What is the tension in the rope if we find an angle $\theta = 10$ degrees, and $F_\perp = 500 \text{ N}$?
5. If the force required to unstuck the vehicle from the mud is 1200 N , does it move?
2. A $20,000 \text{ kg}$ jet fighter lands on an aircraft carrier, moving at 108 km/hr . A tow cable grabs the aircraft and pulls it to a stop in 100 meters . (a) What is the average acceleration? (b) What force does the tow cable exert to stop the jet?